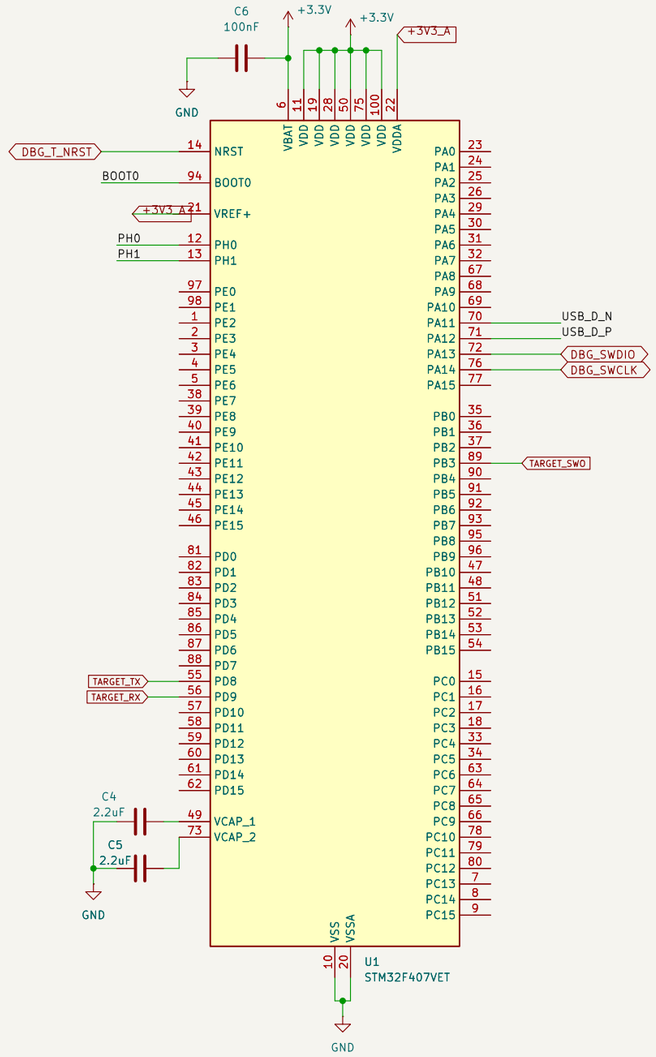
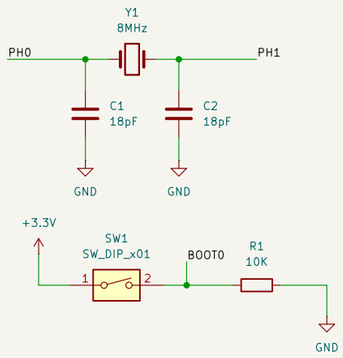
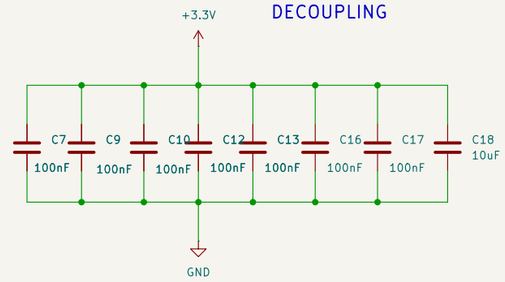


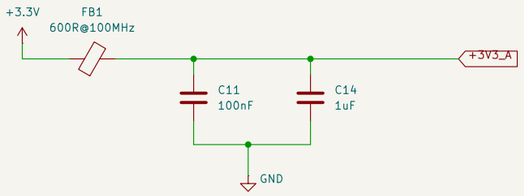
BOOT/RESET/CLOCK



DECOUPLING

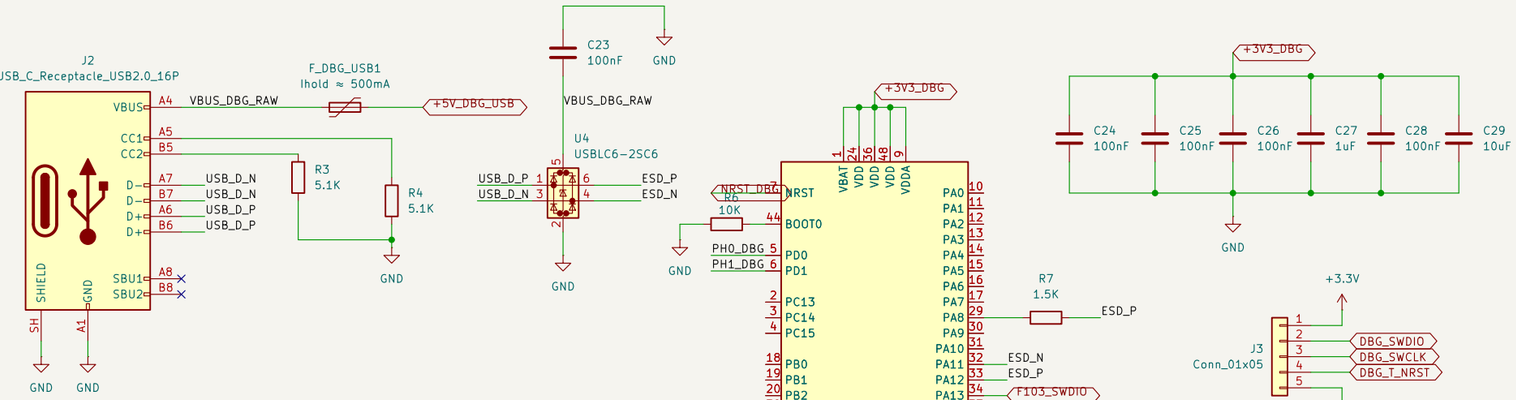


ANALOG SUPPLY



#FLG07
PWR_FLAG

Sheet: /MCU_Core/		
File: STM32F407VET6_Minimum_System.kicad_sch		
Title: STM32F407VET6_KIT		
Size: A3	Date: 2026-10-25	**Rev:**
KiCad E.D.A. 10.0.7		Id: 1/3



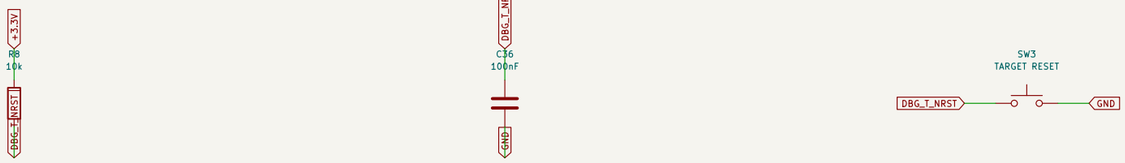
DEBUG ACCESS / ISOLATION — CMSIS-DAP firmware pin mapping must match PB7 / PB8 / PB9.

JP2–4: fit only for onboard debug with BOTH rails powered; remove for external probe or unpowered target.



J4 pin1 / J5 pin1 are voltage references, NOT inputs for a second power supply.

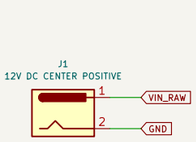
J5: SWO=F407 PB3; TX=PD8, RX=PD9 (USART3). No unverified connection to F103 firmware.

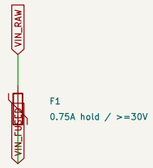


Sheet: /Onboard_Debugger/		
File: onboard_debugger.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 2/3

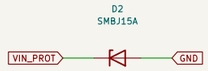
POWER: 12 V regulated bench adapter / optional USB. Not automotive battery input.

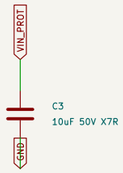
DC protection -> buck 5 V -> diode OR -> buck 3.3 V. USB_TARGET jumper normally OPEN.

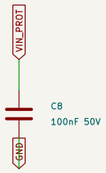




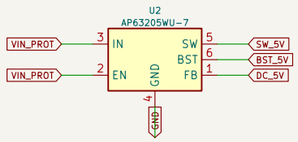


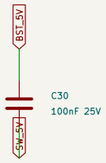


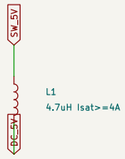


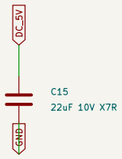


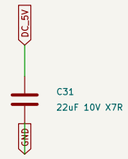
5 V BUCK: AP63205WU-7, datasheet Fig.21 / Table 3. Feedback after inductor, before OR diode.



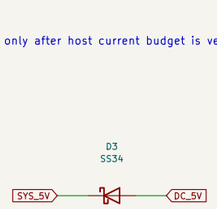


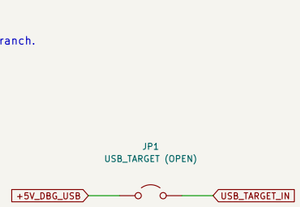


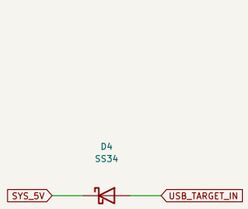


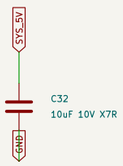


USB supply only after host current budget is verified; F_DBG_USB1 limits the shared USB branch.

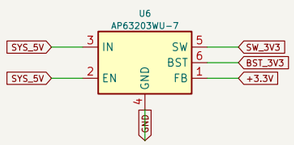


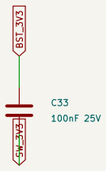


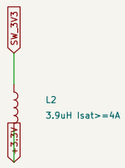


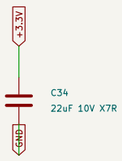


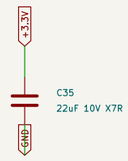
3.3 V BUCK: AP63203WU-7, datasheet Fig.21 / Table 2. Input must remain >=3.8 V.































Sheet: /Power_Management/ File: power_management.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 3/3